

AWS CLOUD MONITORING AND ALERTING SYSTEM

(Using EC2,RDS,Load Balancer,CloudWatch &SNS)

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Course: AWS Cloud Computing

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1. INTRODUCTION:

- The AWS Cloud Monitoring and Alerting System is a cloud-based monitoring project designed to monitor the performance and health of AWS infrastructure resources.
- The project uses Amazon CloudWatch to collect and visualize monitoring metrics from resources such as EC2, RDS, and Elastic Load Balancer. CloudWatch alarms are configured to identify abnormal resource conditions, while Amazon SNS is used to provide notification alerts.
- Amazon SNS (Simple Notification Service) is integrated with the monitoring system to support alert notification. When a configured CloudWatch alarm changes its state, SNS can deliver a notification to the subscribed user.

PROJECT OVERVIEW:

- The project consists of three main monitored infrastructure components: EC2, RDS, and Elastic Load Balancer. CloudWatch acts

as the central monitoring service, while CloudWatch Alarms and SNS provide the alerting mechanism.

The overall monitoring flow is:

AWS Resources → Amazon CloudWatch → CloudWatch Alarms → Amazon SNS → Notification

2. OBJECTIVES:

The main objectives of this project are:

- To monitor the performance and health of AWS infrastructure resources
- .
- To track important metrics of EC2, RDS, and Load Balancer using Amazon CloudWatch.
- To configure CloudWatch Alarms for detecting resource issues.
- To create a centralized CloudWatch Dashboard for easy monitoring.
- To configure Amazon SNS for receiving monitoring alerts
- .
- To improve infrastructure visibility and enable timely response to performance issues.

3. AWS SERVICES USED:

- **Amazon EC2** - Provides the compute server for the application.
- **Amazon RDS** - Provides the managed database service.
- **Elastic Load Balancer** - Distributes incoming application traffic.
- **Target Group** - Manages and monitors registered EC2 targets.
- **Amazon CloudWatch** - Monitors AWS resources and collects performance metrics.

- **Amazon SNS** - Sends notification for CloudWatch alerts.
- **CloudWatch Alarms** - Detects defined resource condition and triggers alerts.

4. Project Architecture:

The project uses Amazon CloudWatch as the central monitoring service for EC2, RDS, and Elastic Load Balancer.

CloudWatch collects the performance metrics of these resources and displays them through graphs and dashboards. CloudWatch Alarms monitor the configured conditions, and Amazon SNS is used to send alert notifications.

This architecture provides a simple and centralized way to monitor AWS resources and receive alerts when required.

EC2 + RDS + Load Balancer



Amazon CloudWatch



CloudWatch Alarms



Amazon SNS



Notification

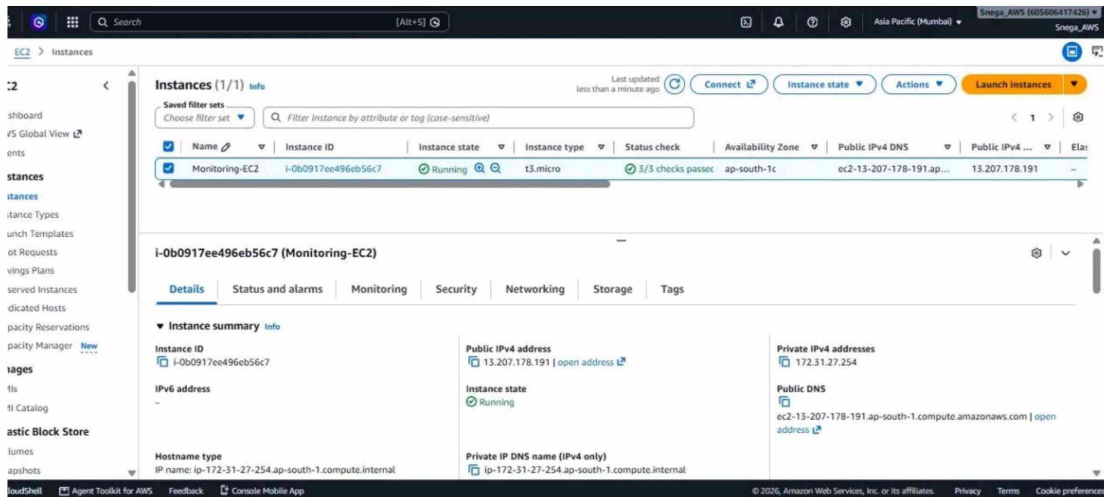
5. EC2 MONITORING:

- Amazon EC2 is used as a compute resource in the project. Amazon CloudWatch is used to monitor the performance and health of the EC2 instance.
- The CPU Utilization metric is monitored to understand the instance workload. A CloudWatch alarm is configured to identify when the CPU utilization reaches the defined threshold.

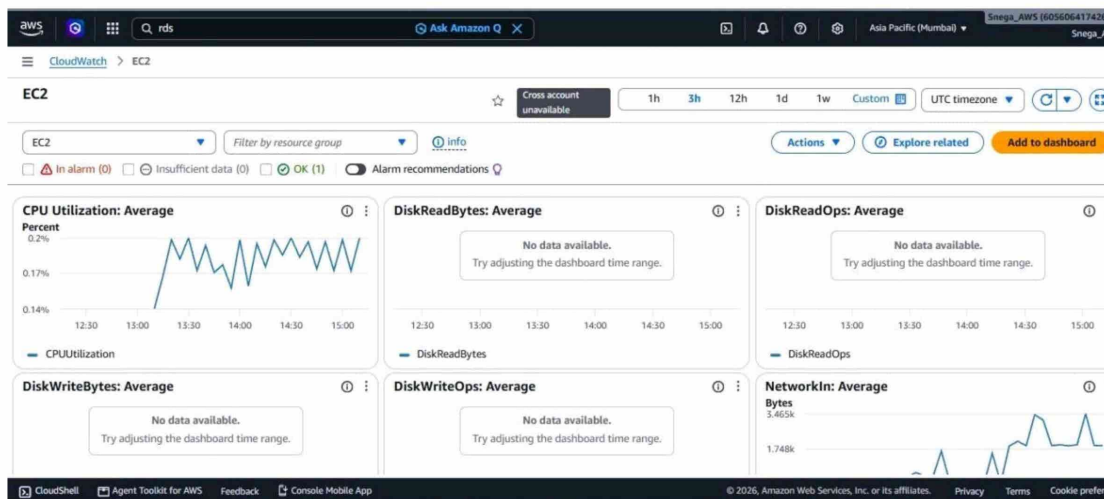
- The EC2 monitoring graph and alarm status are verified to ensure that the instance can be monitored effectively.

Figure 1: EC2 Instance Monitoring and CloudWatch Alarm

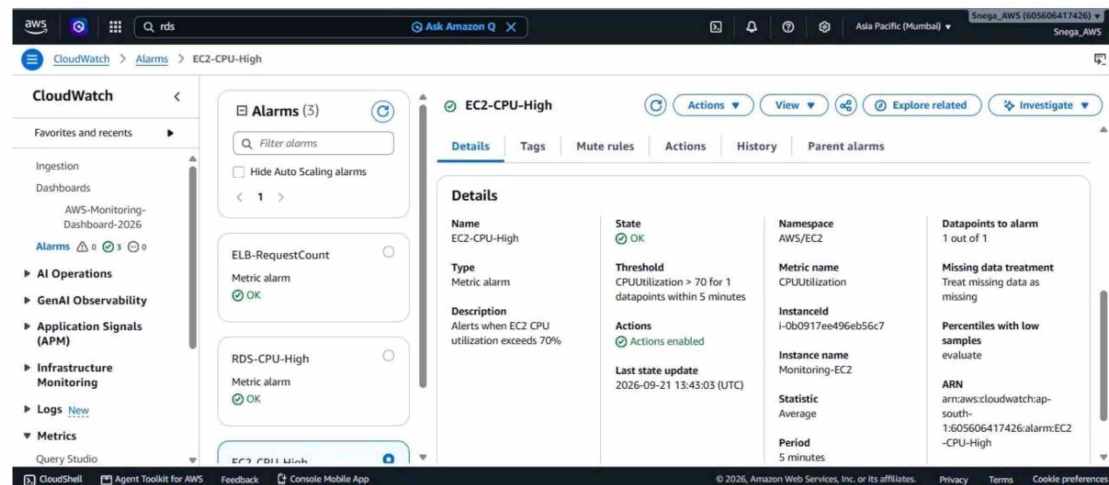
EC2 Instance:



CPU Utilization Monitoring



CloudWatch Alarm:



Result:

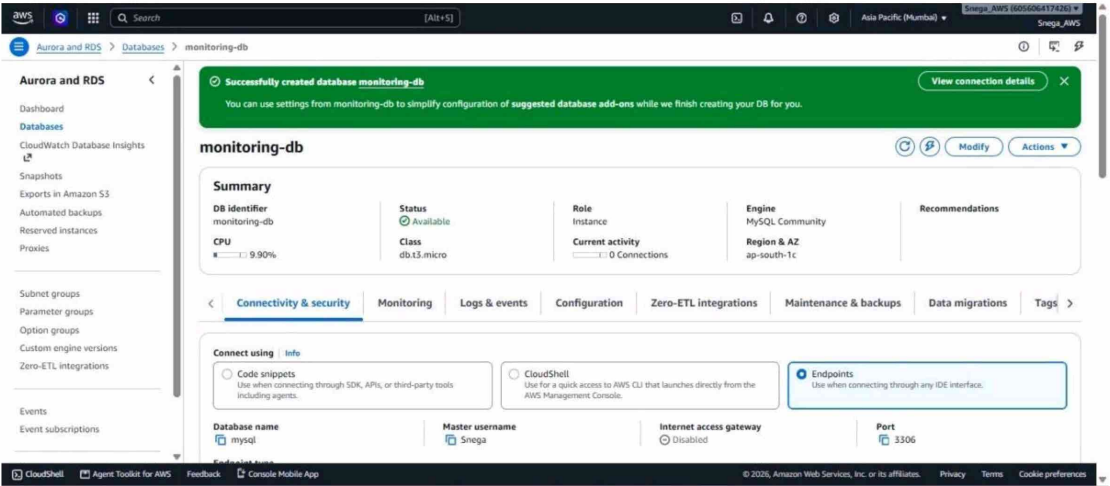
EC2 instance was successfully created and monitored using CloudWatch . CPU utilization was displayed, and the Cloudwatch alarm was configured successfully.

6. RDS MONITORING:

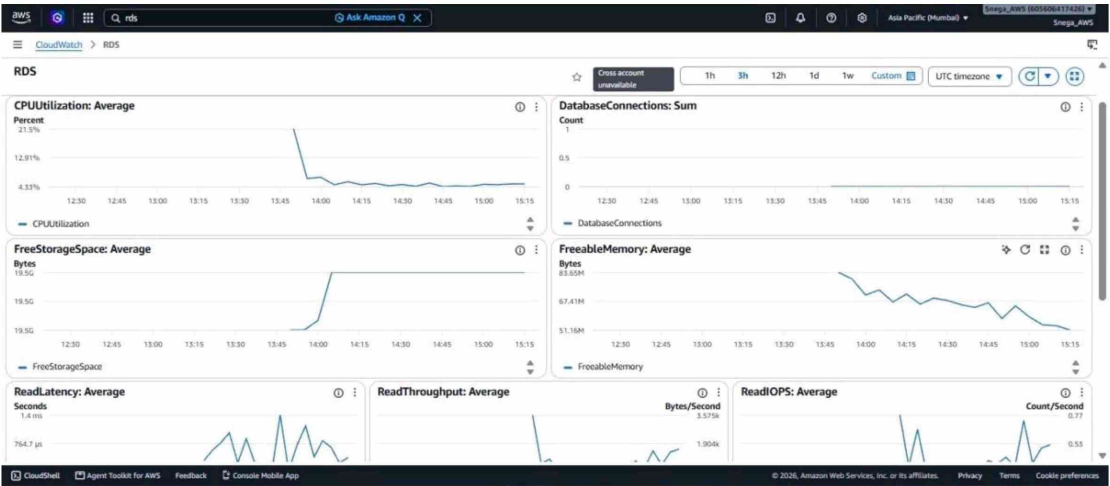
- Amazon RDS is used as the database service in the project. Amazon CloudWatch is used to monitor the performance and health of the RDS database instance.
- Important database metrics are monitored through CloudWatch to observe the resource performance. A CloudWatch alarm is configured to identify the defined performance condition.
- The RDS instance status, monitoring graph, and alarm status are verified to ensure effective database monitoring.

Figure 2: RDS Instance Monitoring and CloudWatch Alarm

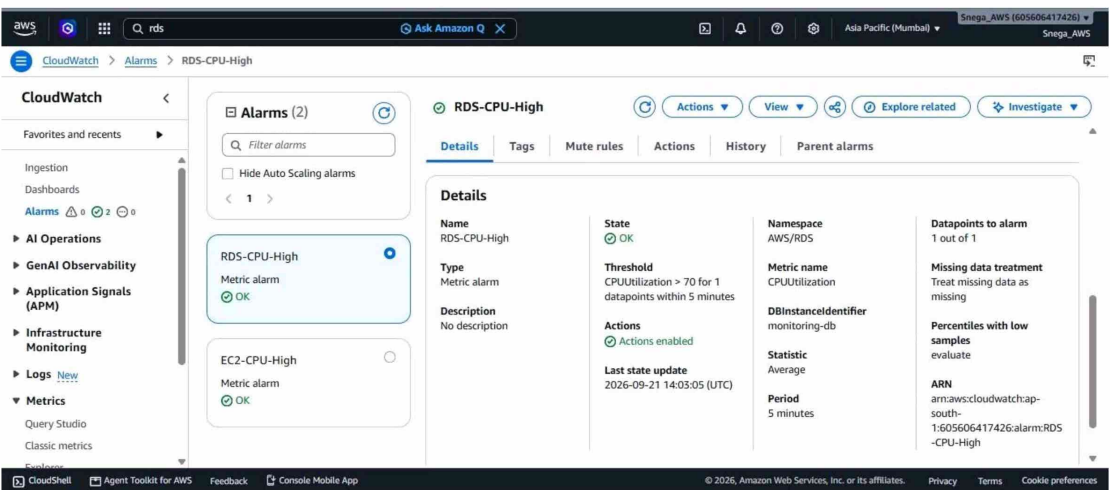
RDS Database:



Monitoring Graph:



CloudWatch Alarm:



Result:

RDS instance was successfully created and monitored using CloudWatch. The database metrics were displayed, and the CloudWatch alarm was configured successfully.

7. LOAD BALANCER and TARGET GROUP MONITORING:

- The Elastic Load Balancer is used to distribute incoming application traffic across the registered targets. The associated Target Group is monitored to check the health status of the registered EC2 instances.
- Amazon CloudWatch is used to monitor the Load Balancer metrics and observe its performance. A CloudWatch alarm is configured to identify the defined monitoring condition.
- The Load Balancer, Target Group health status, monitoring graph, and alarm status are verified to ensure effective traffic and resource monitoring.

Figure 3: Load Balancer, Target Group and CloudWatch Monitoring

Load Balancer:

The screenshot displays the AWS Management Console interface for the 'Monitoring-alb' Elastic Load Balancing (ELB) instance. The console shows the instance is 'Active' and provides details such as the VPC, Availability Zones, and Load balancer IP address type. The 'Details' section includes the Load balancer type (Application), Scheme (Internet-facing), Hosted zone (ZP97RAFLXTNZK), and Load balancer ARN. The 'DNS name info' section shows the DNS name 'Monitoring-alb-53737964.ap-south-1.elb.amazonaws.com (A Record)'. The console also features a sidebar with navigation options like 'Elastic Block Store', 'Network & Security', 'Load Balancing', and 'Auto Scaling'. The top navigation bar includes the AWS logo, search bar, and account information.

Monitoring-alb			
Details			
Load balancer type Application	Status Active	VPC vpc-0fa2f0caee480170	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZP97RAFLXTNZK	Availability Zones subnet-0d12af3454a5711b1 (ap-south-1c) subnet-00b5a27b43fb85e86 (ap-south-1a) subnet-0a11-a21	Date created September 21, 2026, 19:44 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:ap-south-1:605606417426:loadbalancer/app/Monitoring-alb/e3a64042624f59a0		DNS name info Monitoring-alb-53737964.ap-south-1.elb.amazonaws.com (A Record)	

Target Group

The screenshot shows the AWS Management Console interface for a Target Group. The left sidebar contains navigation links for EC2, Instances, Images, and CloudShell. The main content area displays the details for 'MyTargetGroup'.

MyTargetGroup Details:

- Target type:** Instance
- Protocol : Port:** HTTP: 80
- Protocol version:** HTTP1
- VPC:** vpc-0fa2f0eaeef480170
- IP address type:** IPv4
- Load balancer:** None associated

Targets:

Target ID	Health	Unhealthy	Unused	Initial	Draining
1	Healthy	0	0	0	0

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Last fetched seconds ago

CloudWatch Monitoring

The screenshot shows the AWS CloudWatch console for an Application ELB. The dashboard displays several metrics over a 1-hour period.

Application ELB Metrics:

- RequestCount: Sum** (Count): Shows a line graph with two data series: 'app/Monitoring-alb/4bb2fefe76d368ce' and 'app/Monitoring-alb/e3a64042624f59a0'.
- HTTPCode_ELB_5XX_Count: Sum** (Count): No data available.
- ActiveConnectionCount: Sum** (Count): No data available.
- ClientTLSNegotiationErrorCount: Sum** (Count): No data available.
- ConsumedLCUs: Average** (Count): No data available.
- HTTP_Fixed_Response_Count: Sum** (Count): No data available.

CloudWatch Alarm

The screenshot shows the AWS CloudWatch console for an alarm named 'ELB-RequestCount'.

ELB-RequestCount Alarm Details:

- Name:** ELB-RequestCount
- State:** OK
- Namespace:** AWS/ApplicationELB
- Datapoints to alarm:** 1 out of 1
- Type:** Metric alarm
- Threshold:** RequestCount > 0 for 1 datapoints within 5 minutes
- Metric name:** RequestCount
- Description:** No description
- Actions:** Actions enabled
- LoadBalancer:** app/Monitoring-alb/e3a64042624f59a0
- Last state update:** 2026-09-21 15:02:36 (UTC)
- Statistic:** Sum
- Period:** 5 minutes
- Missing data treatment:** Treat missing data as missing
- Percentiles with low samples:** evaluate
- ARN:** arn:aws:cloudwatch:ap-south-1:605606417426:alarm:ELB-RequestCount

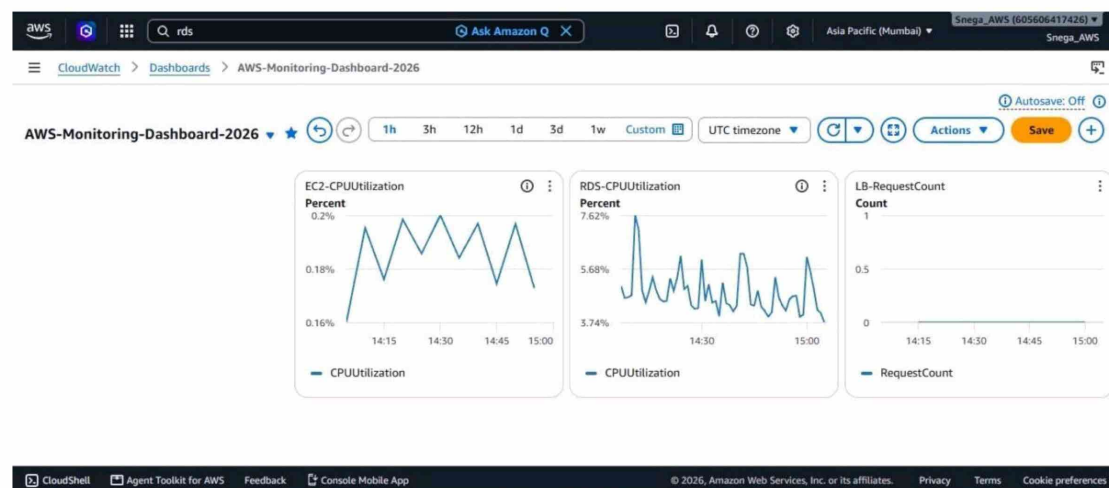
Result

The Load Balancer was successfully configured and monitored using CloudWatch. The Target Group health was verified, and the CloudWatch alarm was configured successfully.

8. CloudWatch Dashboard:

- A centralized Amazon CloudWatch Dashboard is created to display the monitoring metrics of the AWS resources in a single view.
- The dashboard provides a convenient way to monitor the performance of EC2, RDS, and Elastic Load Balancer. It helps to view the resource metrics and identify performance changes quickly.
- This centralized dashboard improves monitoring visibility and makes it easier to review the overall health of the AWS infrastructure.

Figure 4: Amazon CloudWatch Monitoring Dashboard

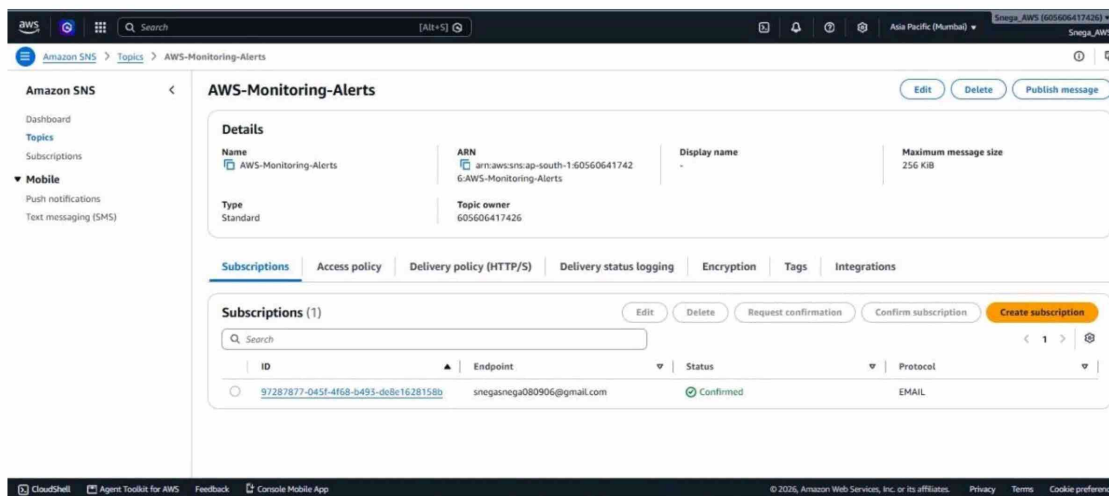


9. SNS NOTIFICATION:

- Amazon SNS is used to send notifications when a CloudWatch alarm changes its state.

- An SNS topic and email subscription are configured to receive monitoring alerts. The CloudWatch alarms are connected with SNS so that notifications can be sent when the configured alarm condition is triggered.
- The SNS subscription is confirmed to ensure that the notification service is ready to deliver CloudWatch alerts.

Figure 5: Amazon SNS Topic and Confirmed Subscription



Result

The SNS topic and email subscription were successfully configured. The Subscription was confirmed and ready to receive CloudWatch alerts.

10. Testing and Verification:

The configured AWS monitoring components were tested and verified to ensure that the monitoring system works as expected.

The following components were successfully verified:

- EC2 instance monitoring and CPU utilization graph.
- RDS instance status and monitoring graph.
- Load Balancer and Target Group health status.
- CloudWatch monitoring dashboard.
- CloudWatch alarms for the monitored resources.
- SNS email subscription and notification configuration.

11. FUTURE ENHANCEMENTS:

- Integrate AWS Lambda for automated actions.
- Add monitoring for more AWS services.
- Implement advanced CloudWatch dashboards.
- Add centralized log monitoring using CloudWatch Logs.
- Configure automated monitoring reports.
- Add AWS cost monitoring and alerts.

12. CONCLUSION:

- The AWS Cloud Monitoring and Alerting System project successfully demonstrated the monitoring of AWS resources using Amazon CloudWatch. EC2, RDS, and Elastic Load Balancer metrics were monitored through CloudWatch dashboards and alarms.
- The integration of Amazon SNS provided a notification mechanism for monitoring alerts. This project provided practical knowledge of AWS resource monitoring, metric analysis, alarm configuration, dashboard creation, and notification management.
- Overall, the project demonstrates how AWS monitoring services can be used to improve infrastructure visibility and support timely identification of resource performance issues.